
Multi-modal Emotion Recognition Using Transformers with Cross-Attention Mechanisms

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Abstract

This project investigates multimodal emotion recognition using a Transformer-based architecture with cross-attention mechanisms. Building on prior works such as MulT, MISA, and MAG-BERT, our model fuses audio and visual modalities to capture expressive human emotions. We use the CMU-MOSEI dataset, extracting COVAREP features from audio and FACET 4.2 features from video, and construct a balanced subset to mitigate class imbalance. Each modality is processed through a dedicated Transformer encoder to model temporal dependencies, followed by cross-attention layers that enable inter-modal interaction and contextual alignment. The fused representations are temporally aggregated and passed through a fully connected layer for emotion prediction. This architecture is designed to learn both intra-modal and inter-modal relationships, enhancing robustness to noisy or missing cues. Our approach demonstrates the potential of attention-based fusion for capturing fine-grained emotional expressions from speech and facial dynamics.

1 Introduction

Emotion recognition from speech is inherently multi-modal, as human affect is expressed not only through vocal tone but also through facial expressions and nonverbal cues. While uni-modal systems based solely on text or audio features have achieved some success, they often miss the complementary information present across modalities. Recent advances in multi-modal learning have shown that integrating multiple modalities significantly enhances emotion classification performance.

Our work builds on this foundation by focusing on emotion recognition using audio and visual modalities from the CMU-MOSEI dataset. We exclude textual data to isolate nonverbal cues and better understand how affective information is conveyed through tone and expression. Inspired by prior multi-modal architectures such as the Multi-modal Transformer (MulT) [1], MISA [2], and MAG-BERT [3], we adopt a transformer-based model with cross-attention layers to enable dynamic interaction between modalities. Each modality is encoded through a dedicated transformer encoder, allowing the model to learn temporal dependencies within the audio and visual streams before combining them through cross-attention for emotion classification.

This design allows our system to model both modality-specific patterns, such as pitch variations or facial dynamics, and their interrelations, leading to a more expressive representation of emotional states. The project also addresses practical challenges such as class imbalance, feature synchronization, and normalization to ensure reliable training and fair emotion distribution across the dataset.

2 Background Information and Related Works

Emotion recognition from speech is inherently multi-modal, as vocal tone, spoken words, and facial expressions jointly convey affective states. Though uni-modal approaches which depended solely on text-transcriptions or audio features have had success, more recent works have illustrated that combining multiple modalities notably improves sentiment and emotion classification performance. Our project builds on this multi-modal foundation by focusing on speech-based emotion recognition using audio (frequency-based features) and visual features.

The CMU-MOSEI dataset has become one of the most widely used benchmarks for this task. It consists of short, annotated video segments from YouTube where each utterance is labeled with sentiment and six emotion intensities. In our project, we focus on the audio-visual subset of MOSEI to better understand nonverbal expressions of emotion, excluding text to avoid linguistic bias and isolate acoustic and facial cues. For audio, CMU-MOSEI dataset uses COVAREP toolkit extracts 74-dimensional prosodic and spectral features. For the visual stream, FACET 4.2 produces 35-dimensional features per frame, representing facial Action Units (AUs), gaze direction, and expression intensities.

The Transformer architecture [4] revolutionized sequence modeling by replacing recurrence with a self-attention mechanism capable of modeling long-range dependencies and parallel training. Transformer-based fusion models have rapidly become state of the art for multi-modal emotion recognition. The Multi-modal Transformer (MulT) [1] introduced directional cross-modal attention to learn temporal alignments between modalities, achieving large gains over early and late-fusion baselines. In multi-modal learning, attention mechanisms are particularly effective because they can dynamically assign weights to information from different time steps or modalities. Cross-attention extends this idea by allowing one modality to attend to another, enabling fine-grained alignment. Other works, such as Audio-Visual Temporal Transformer with Cross-Attention (AVT-CA) [5], further highlighted the value of dynamic attention to synchronize audio and facial streams.

3 Ethical Considerations

Our project, while grounded in publicly available datasets like CMU-MOSEI, involves modalities; audio, text, and video, that carry inherent privacy concerns, especially given their capacity to reveal identity and emotional states. Although the dataset was collected with consent, deploying emotion recognition systems more broadly could raise issues around surveillance, consent, and autonomy. Moreover, such models risk reinforcing biases, as emotional expression varies across cultures, genders, and dialects, some groups may be underrepresented in training data. This can lead to misclassifications or unfair treatment in real-world settings.

Misuse of emotion recognition, such as in secret monitoring or behavioral profiling, also remains a serious concern. Example of such applications may result in individuals being evaluated or judged without their knowledge, potentially leading to discrimination, emotional manipulation and or violations of personal dignity, particularly, if the model’s evaluations are treated as objective truth rather than probabilistic predictions.

Another critical limitation of emotion recognition systems is their vulnerability to intentional manipulation. Since the model infers emotional states from observable acoustic and visual cues, individuals can easily alter their voices or facial expressions to influence the model’s predictions. For instance, a person may speak with a cheerful tone or smile deliberately to appear “happy,” even when their genuine emotional state differs. This susceptibility to deceptive behaviors raises concerns about robustness and reliability in real-world applications, especially in contexts such as hiring, security screening, or psychological assessment. Future work should explore adversarial robustness and multi-modal consistency checks to mitigate these risks.

4 Data

4.1 Dataset Inspection

We use the CMU-MOSEI dataset [6] — a large-scale multi-modal corpus created by Carnegie Mellon University that contains over 23,500 annotated monologue YouTube video segments from more

than 1,000 speakers across diverse topics. Each segment is annotated for sentiment and six discrete emotions, and includes synchronized audio, visual, and text modalities. The dataset is approximately gender-balanced and covers a wide range of speaking styles, accents, and environments, which makes it a robust benchmark for multi-modal emotion recognition.

4.2 Data Used

In this work, we focus on the audio and visual modalities, as they are the most expressive of affective states. The textual modality is excluded to isolate the contribution of nonverbal signals.

Table 1: Audio, video features extractions

Modality	Feature Extractor	Dimensionality
Audio	COVAREP	74 features per frame
Video	FACET 4.2	35 features per frame

4.3 Data Labels

Each segment is accompanied by a label vector of seven components: the first element represents sentiment (ranging from -3 to $+3$), while the remaining six correspond to the intensity ($0 - 3$) of happiness, sadness, anger, disgust, surprise, and fear respectively.

4.4 Data Analysis and Dataset Characteristics

Emotion distribution analysis reveals strong class imbalance in CMU-MOSEI, with over 12,000 segments labeled as happiness but only about 1,900 labeled as fear. This imbalance can bias models toward majority emotions and motivates the need for data balancing.

Additionally, the emotion labels are not one-hot vectors but continuous intensity values (ranging from 0 to 3 for each emotion). These scores represent the average subjective ratings from human annotators, which introduces inherent noise and uncertainty into the ground truth, making precise emotion classification more challenging.

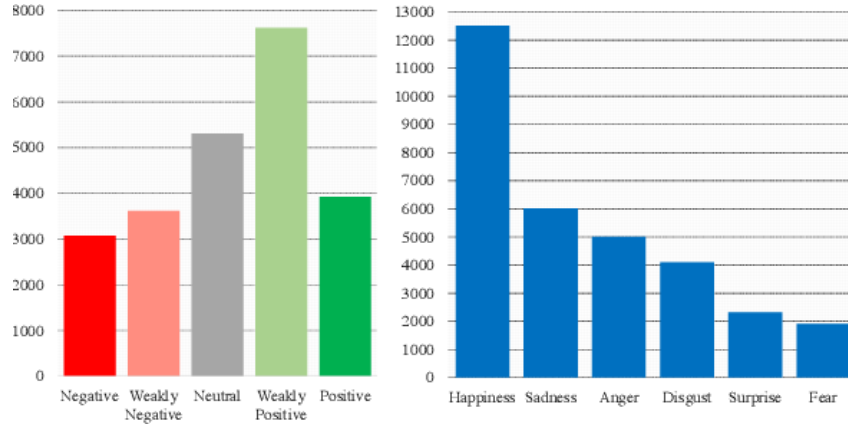


Figure 1: Sentiment distribution (left) and emotion label distribution (right) in the CMU-MOSEI dataset [6].

4.5 Data Cleaning and Preprocessing

- **Balancing:** The dataset is highly imbalanced: *happiness* is the most frequent emotion, whereas *fear* has only about 1,900 samples. To mitigate bias, we subsample approximately 1,900 instances per emotion, producing a balanced subset.
- **Feature Synchronization:** Video quality and frame rates vary across samples. To standardize input, we extract COVAREP features from audio at 100 Hz using 25 ms windows with 10 ms hops,

and **FACET 4.2** features from video at 30 fps. Both modalities are temporally aligned to ensure frame-level correspondence.

- **Normalization:** Each feature dimension is standardized via z-score normalization (mean 0, variance 1) computed over the training set to ensure stable optimization.
- **Sequence Formatting:** Each utterance is truncated or zero-padded to 500 frames (approximately 5 s), corresponding to the average segment duration in CMU-MOSEI. This enables efficient batch processing and parallel computation on GPUs.
- **Storage Format:** The preprocessed data are stored as NumPy arrays and serialized into .npz files for reproducible experiments and efficient loading during training.

5 Model architecture

We propose a multi-modal Transformer-based architecture for emotion recognition that jointly models audio and visual modalities. Motivated by recent studies demonstrating the effectiveness of Transformer and cross-attention mechanisms in multi-modal emotion classification [5], we first extract speech and facial features from raw audio and video inputs. Two modality-specific Transformer encoders capture temporal dependencies within each modality by self-attention, while cross-attention layers enable inter-modal interaction [4],[1]. The fused representations are temporally pooled to pick on the moment that affect the prediction the most [7] and passed through a fully connected layer for emotion prediction, allowing the model to effectively learn both intra-modal and inter-modal relationships crucial for robust emotion understanding.

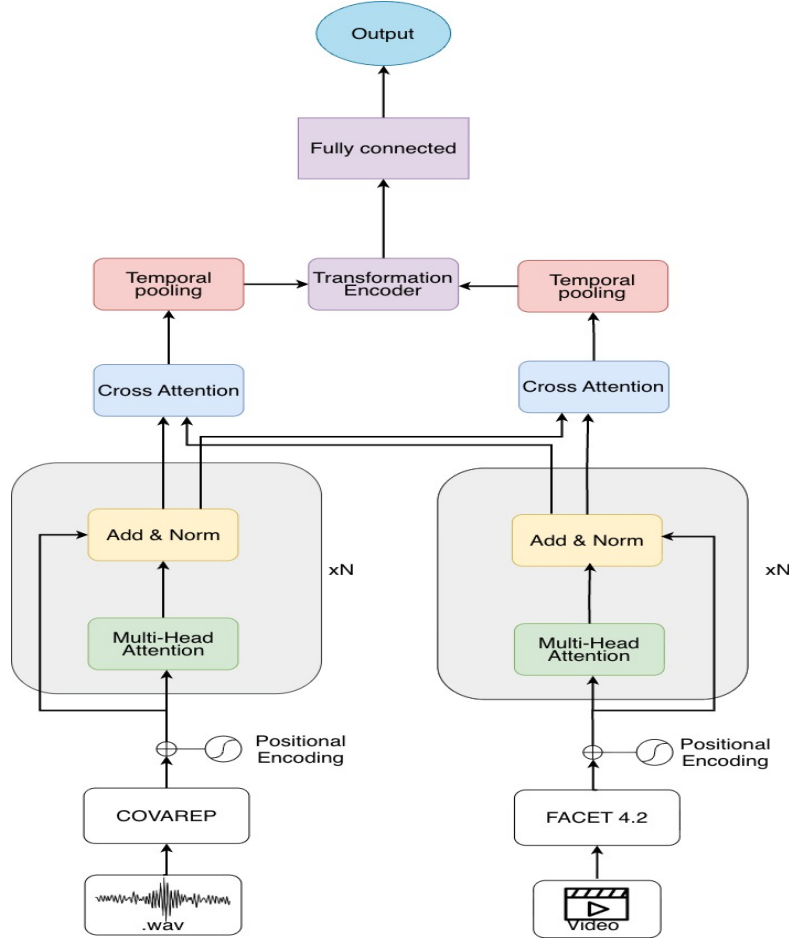


Figure 2: Overview of the multi-modal emotion recognition model combining audio and video inputs.

6 Work Division

The team meets every Tuesday and Friday between 6:30pm - 7:30pm ET for planning, progress review, and discussion of blockers. Each meeting lasts approximately 30 minutes to an hour, with additional asynchronous collaboration via shared documents, version control, with Discord being our main communication channel. The workload is divided to balance contributions across members while ensuring clarity of responsibilities. Estimated weekly time commitments are indicated for each role, along with notes on potential coordination points.

Task Assignment:

- **Data Cleaning and Preprocessing (Zuhayr, Abuzar):** Balancing the dataset, synchronizing audio and visual features, normalizing inputs, and formatting sequences. Estimated 5 hours/week. Coordination with the Architecture team ensures data meets model input requirements.
- **Model Architecture Tuning and Training (Zuhayr, Khoi):** Implementing Transformer encoders, cross-attention layers, and training loops. Experimenting with architectural variations and training on preprocessed data. Estimated 6 hours/week. Requires communication with Data Cleaning and Validation teams to iterate on data and model performance.
- **Validation and Hyperparameter Optimization (Shah, Khoi):** Conducting validation runs and tuning hyperparameters such as learning rate, batch size, attention heads, and dropout. Estimated 4 hours/week. Collaboration with Architecture team ensures application of optimal settings.
- **Evaluation and Testing (Abuzar, Shah):** Running evaluation protocols, computing metrics, generating plots, and analyzing results. Estimated 5 hours/week. Coordination with Validation team ensures testing uses finalized model configurations.
- **Report Preparation (Abuzar, Khoi, Shah, Zuhayr):** Documenting methodology, experiments, and results. Maintaining consistency in figures, tables, and formatting. Estimated 4 hours/week, with increased effort near deadlines. Requires input from all teams to integrate results and analyses.

The listed members are primarily responsible for each task, but all team members contribute through discussion, review, and collaborative input.

References

- [1] Y.-H. H. Tsai, S. Bai, P. P. Liang, J. Z. Kolter, L.-P. Morency, and R. Salakhutdinov, “Multimodal transformer for unaligned multimodal language sequences,” 2019.
- [2] D. Hazarika, R. Zimmermann, and S. Poria, “Misa: Modality-invariant and -specific representations for multimodal sentiment analysis,” in *Proceedings of the 28th ACM International Conference on Multimedia*, MM ’20, (New York, NY, USA), p. 1122–1131, Association for Computing Machinery, 2020.
- [3] W. Rahman, M. K. Hasan, S. Lee, A. Zadeh, C. Mao, L.-P. Morency, and E. Hoque, “Integrating multimodal information in large pretrained transformers,” 2020.
- [4] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” 2023.
- [5] J. D. P. R, S. Venkatraman, V. Sharma, S. Malarvannan, and M. Narendra, “Multimodal emotion recognition using audio-video transformer fusion with cross attention,” 2025.
- [6] A. Zadeh, P. P. Liang, S. Poria, E. Cambria, and L.-P. Morency, “Multimodal language analysis in the wild: CMU-MOSEI dataset and interpretable dynamic fusion graph,” in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, (Melbourne, Australia), pp. 2236–2246, Association for Computational Linguistics, 2018.
- [7] Z. Lin, M. Feng, C. N. dos Santos, M. Yu, B. Xiang, B. Zhou, and Y. Bengio, “A structured self-attentive sentence embedding,” 2017.